

Week 10 Lab Report

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Question 3

We throw a fair die n times and study the distribution of the mean of these n throws. Set $n = 2$ in this question.

Part a)

Use the function `sample()` to simulate n throws. Repeat this $r = 10000$ times, compute the mean of each repetition, and store the results in `sm`.

```
set.seed(0503)

r <- 10000
n <- 2
sm <- rep(0, r)
for (i in 1:r)
{
  x <- sample(1:6, n, replace=TRUE)
  sm[i] <- mean(x)
}

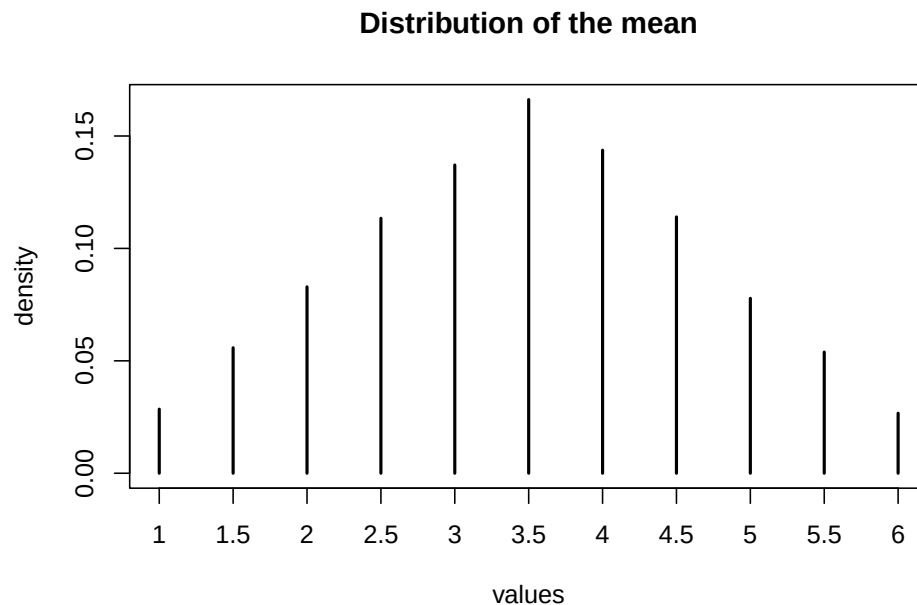
head(sm)
```

```
## [1] 4.5 2.0 4.0 5.0 2.5 3.0
```

Part b)

Plot the distribution of `sm` and comment on the shape of the distribution. Does the CLT apply here?

```
plot(table(sm)/r,
      xlab="values",
      ylab="density",
      main="Distribution of the mean")
```



This is not a bell-shaped distribution. In fact the distribution is triangular. The sample size of $n = 2$ is much too small for CLT to apply.

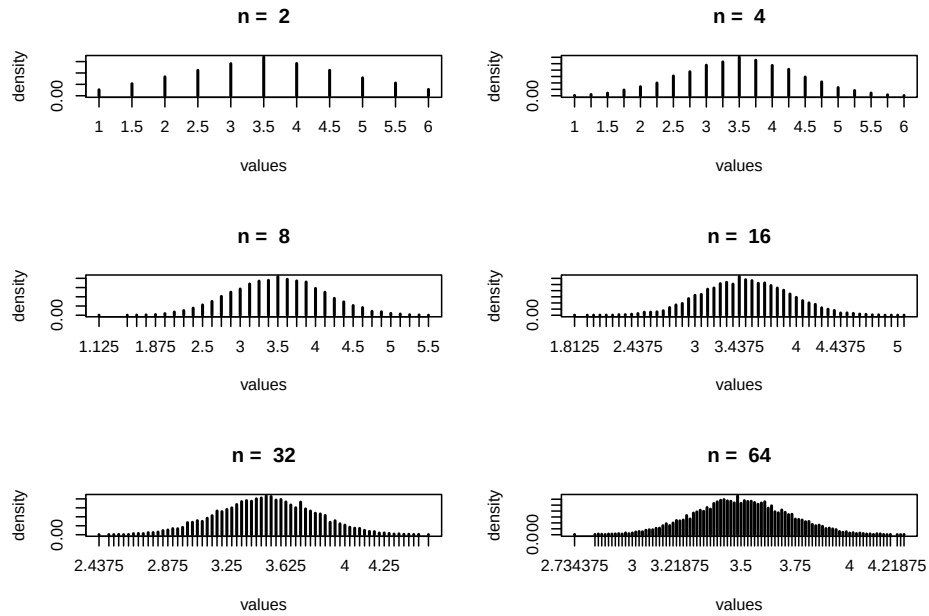
Part c)

Repeat the above procedure for $n = 4, 8, 16, 32$ and comment on how the shape of the distribution changes, relating to the CLT.

```
r <- 10000
n <- c(2, 4, 8, 16, 32, 64)
sm <- rep(0, r)

par(mfrow=c(3, 2))

for (j in n)
{
  for (i in 1:r)
  {
    x <- sample(1:6, j, replace=TRUE)
    sm[i] <- mean(x)
  }
  plot(table(sm)/r,
       xlab="values",
       ylab="density",
       main=paste("n = ", j))
}
```



As the number of coin tosses increases (i.e. an increased number of observations in each repetition), the distribution becomes more and more symmetrical and bell-shaped, resembling a normal curve by the time $n = 16$. This is in line with what the CLT states, which is that as the sample size gets larger, the distribution of sample means $\bar{x}$ approximates a normal distribution, regardless of the population's distribution.